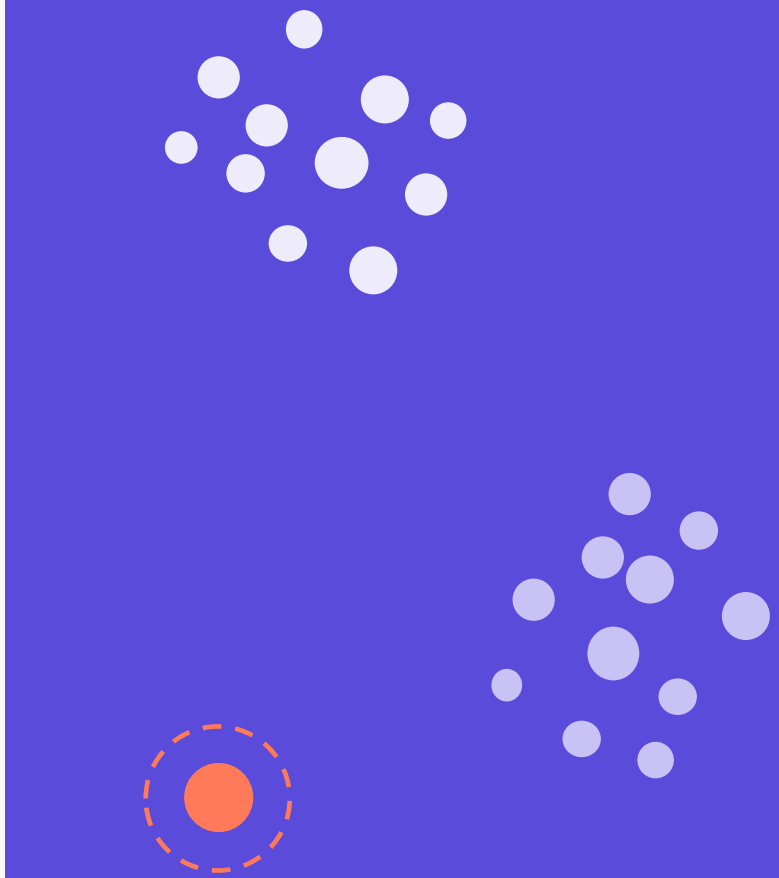
Week 8 DAML Workshop

Autoencoders, DEC clustering and anomaly detection



NOTEBOOK OVERVIEW

This week's notebook at a glance

As with previous weeks, fill in every `## FINISH_ME ##` in the code and in the questions.

SECTION 1



Data handling

Hands-on with passing data between **numpy**, **Pandas** and **PyTorch**

SECTIONS 2–4



Autoencoders

Build, train and analyze an **Auto-Encoder** and a **Variational Auto-Encoder**

SECTION 5



DEC clustering

Train a **DEC model** on your pre-trained AE, then analyze its output

SECTION 6



Anomaly hunt

Use your AE and a **smiley-face image** to spot anomalies. Open-ended.

SECTION 5

Train a DEC model on top of your pre-trained autoencoder

Training a DEC model reliably requires 2 extra things

INPUT



Pre-trained AE

DEC models require a pre-trained AE as input

EXTRA 1



Seed the centroids

Seed them correctly before training starts

EXTRA 2



Follow the training scheme

It’s complex: training technically goes through phases

OUTPUT



Trained DEC model

Its output is what you’ll analyze next

SECTION 5

Spend your time analyzing the DEC output, not the training code

LOOK THROUGH



Training material

- A lot of the material for training the model is **provided as-is** for you to look through
- A complete understanding **isn't a hard requirement**

YOUR FOCUS



"I'm more interested in you taking the output from a trained DEC and **analyzing this with plots and graphs.**"

SECTION 6

Relatively open-ended

Use your autoencoder to find anomalies in a smiley face



Your pre-trained AE

+



The smiley-face image provided





Anomaly detection



Examples in the lecture

There's example material relating to this in this week's lecture



Plot it your way

How exactly you choose to plot this is up to you: `numpy`, `pandas`, `tensors`, whatever you feel most comfortable with